Not All Models Are Inflated: Profiling Contamination Dose-Response Across Model Families

Anonymous ACL submission

Abstract

Benchmark contamination is widely assumed to inflate language model scores, yet its actual effect has never been systematically measured across model families. We perform controlled SFT injection at five dosage levels into six models (7B–14.7B) from five open-source families and discover a spectrum of dose-response patterns, not uniform inflation. Two patterns replicate across independent families: Immune (Phi-4, LLaMA-3.1) and Benefit (statistically significant for Gemma-2, BCa 95% CI [+1.49, +3.14] pp; borderline for OLMo-2 and Falcon3). Collapse and V-recovery are each observed in one family (Qwen2.5) across three seeds but remain preliminary pending independent replication. Effects are benchmark-conditional: MMLU-targeted contamination improves GSM8K in five of six families, even those harmed on MMLU. Difference-in-differences analysis detects no item-specific memorization; a volume-matched control attributes 45% of Benefit to data quantity. ContamCal, our diagnostic framework combining multi-signal exposure mapping with contamination-augmented IRT, enables domain-level contamination diagnosis with credible intervals—revealing that Social Sciences contamination effects are negative while STEM effects are positive, a pattern invisible to aggregate scoring. Out-of-sample validation on LLaMA-3.1-8B confirms that model family characteristics beyond baseline ability modulate vulnerability. Model-specific dose-response profiling is necessary for reliable benchmark interpretation.

1 Introduction

Benchmark scores drive model selection and research priorities for large language models, yet training data contamination is pervasive: trillion-token web crawls routinely ingest benchmark items (Oren et al., 2024), and practitioners cannot determine how much of a reported score reflects genuine capability versus memorization (Hendrycks et al., 2021; Chang et al., 2024). Despite twenty mitigation strategies (Sun et al., 2025), contamination-free benchmarks (White et al., 2024), and dynamic regeneration (Dong et al., 2024), a fundamental question remains: what is the actual effect of contamination on aggregate scores? The implicit assumption is uniform inflation, but this has never been tested across model families.

We perform controlled SFT injection (LoRA, five dosage levels) into six models spanning five independent families from five labs (Alibaba, Microsoft, Allen AI, Google, TII) and discover a spectrum of dose-response patterns (Figure 1)—from immunity (Phi-4, ±0.5pp) and benefit (+3–4pp, three families) to collapse (Qwen-14B, −9 to −25pp across seeds).

Across six models, three lower-baseline models benefit (+3.2 to +4.2pp), one is immune (Phi-4), and the remaining two (both Qwen2.5) show harm—with the 14B model 5× more fragile than 7B. These effects are benchmark-conditional: the same model can collapse on MMLU (−9.4pp) yet gain on GSM8K (+13pp). ContamCal, our diagnostic framework combining multi-signal exposure mapping with contamination-augmented IRT (2PL+γ_k), enables this characterization.

Our contributions:

1. **Dose-response profiles with cross-seed and cross-benchmark validation.** Controlled injection across five families reveals two replicated patterns (Immune, Benefit) and two preliminary Qwen2.5-specific patterns (Collapse, V-recovery). Out-of-sample validation on LLaMA-3.1-8B confirms generalizability. Cross-benchmark transfer reveals directional reversal on GSM8K for three families. Multi-seed evaluation ($n=3$) shows direction is reproducible but magnitude is seed-sensitive.

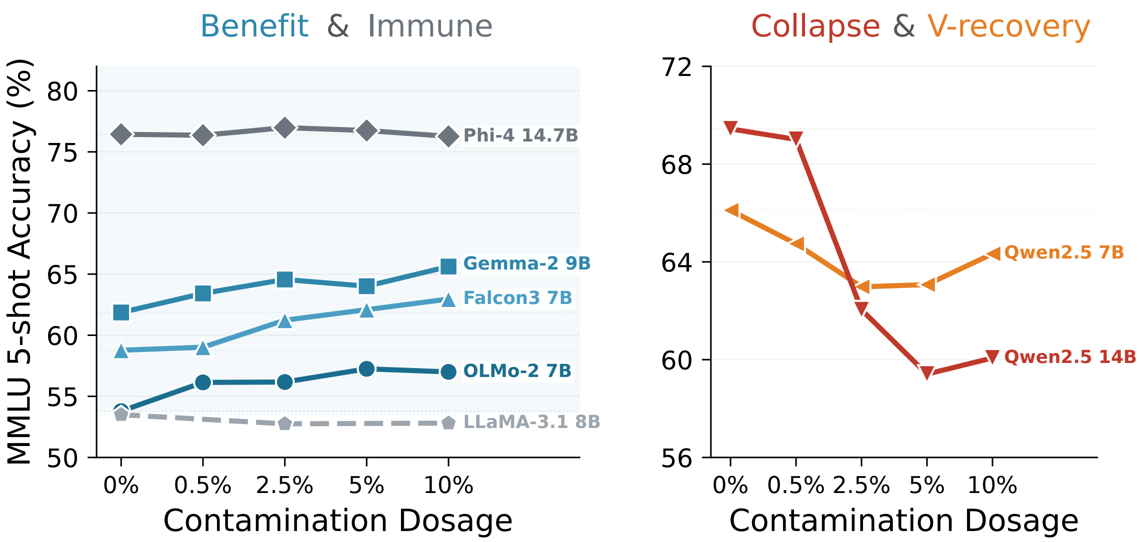

Figure 1: MMLU accuracy vs. contamination dosage (5-shot, seed=42; LLaMA-3.1 at seed=0). Immune and Benefit replicate across families; Collapse and V-recovery are preliminary (Qwen2.5-specific). Cross-seed stability in Table 14.

2. **Domain-level contamination diagnosis via IRT.** ContamCal embeds multi-signal RRF exposure scores in a contamination-augmented IRT model (2PL+γ)—an application of standard explanatory IRT (De Boeck and Wilson, 2004) to LLM contamination—enabling domain-level diagnosis with credible intervals. Domain analysis reveals opposing contamination directions (Social Sciences $\gamma_d < 0$, STEM $\gamma_d > 0$) invisible to aggregate scoring. No item-specific memorization is detected via difference-in-differences (MDES 1.25–2.05pp).

3. **Diagnostic pipeline requirements and signal failure.** Min-K%++ (Zhang et al., 2025) exhibits directional reversal at 0.5B (paired Cohen's $d = -1.98$ at d100). RRF-based multi-signal fusion is essential: a single-signal pipeline produces severe model misfit (PPP < 0.001). Simulation establishes γ recovery requires $J \geq 50$ at AUC ≥ 0.55, above current practice ($J = 8$–30).

Code and data are available at https://github.com/anonymous-nlp-2026-2/contamcal-irt-calibration.

2 Related Work

Contamination detection and mitigation. Detection methods flag contaminated items via token-level surprisal (Shi et al., 2024; Zhang et al., 2025), accuracy gaps on paraphrases or rephrased items (Dekoninck et al., 2024; Yang et al., 2023), kernel divergence (Choi et al., 2025), completion probes (Dong et al., 2024; Golchin and Surdeanu, 2024), and parallel test sets (Zhang et al., 2024); see Chen et al. (2025); Singh et al. (2024); Sainz et al. (2023) for surveys. Balloccu et al. (2024) document contamination malpractices in closed-source LLMs; Bordt et al. (2025) argue that contamination effects are often negligible in practice. Earlier work characterized contamination as memorization (Carlini et al., 2023; Magar and Schwartz, 2022; Li and Flanigan, 2024; Xu et al., 2024); Tao et al. (2025) and Kocyigit and Yildirim (2025) address RL post-training. These methods perform binary classification without quantifying effect sizes. On mitigation, Sun et al. (2025) find none of twenty approaches reliably restores clean performance; DCR (Xu et al., 2025) estimates decontaminated scores but without uncertainty quantification. ContamCal consumes detection signals as inputs and produces calibrated ability with credible intervals. Three concurrent lines of work directly address contamination quantification. Han et al. (2026) apply observational DIF to detect contamination without controlled dosage, achieving detection but not dose-response profiling. Schaeffer et al. (2026) perform controlled pretraining-stage injection with larger seed counts ($n = 10$), establishing causal contamination effects at the pretraining level; our $n = 3$ seed design trades statistical power for broader family coverage. Our work complements both: like Schaeffer et al., we use controlled injection for causal identification, but at the SFT stage with precise dosage control; like Han et al., we employ IRT-based modeling, but with ground-truth contamination labels enabling dose-response characterization. Han et al. apply observational DIF to naturally contaminated models; we use AUGMENT-mode SFT injection with explicit dosage control, enabling causal dose-response profiling.

IRT for LLM evaluation. Item Response Theory (Lord, 1980; Hambleton et al., 1991) has gained traction for LLM evaluation (Liang et al., 2023): Rodriguez et al. (2021), TinyBenchmarks (Polo et al., 2024), Growing Pains (Habba et al., 2026), HiBayES (Luettgau et al., 2025), PSN-IRT (Zhou et al., 2026); see Lalor et al. (2024) for a tutorial and Gu et al. (2024) for evaluation standardization. Most directly related, Han et al. (2026) apply group-based DIF (Holland and Wainer, 1993) to quantify item-level contamination effects. ContamCal differs by using continuous exposure covariates (explanatory DIF; De Boeck and Wilson, 2004), providing calibrated ability with credible intervals, and domain-level diagnosis (γ_d); the approaches are complementary (item-level DIF for benchmark curation, model-level calibration for score interpretation). Our γ parameter is formally an instance of explanatory DIF (De Boeck and Wilson, 2004; Eckerly, 2017): the contamination covariate differentially affects response probability, analogous to item compromise models in educational measurement (Sinharay, 2017; Holland and Wainer, 1993; Camilli and Shepard, 1994; Swaminathan and Rogers, 1990). We use "calibration" in the psychometric sense of adjusting scores to approximate uncontaminated performance, distinct from probability calibration (Guo et al., 2017).

3 Method

ContamCal operates in three stages (Figure 2): Stage 1 estimates per-item contamination exposure via training-free signal fusion; Stage 2 embeds these scores in a contamination-augmented IRT model to recover clean ability estimates with credible intervals; Stage 3 classifies dose-response trajectories into canonical patterns.

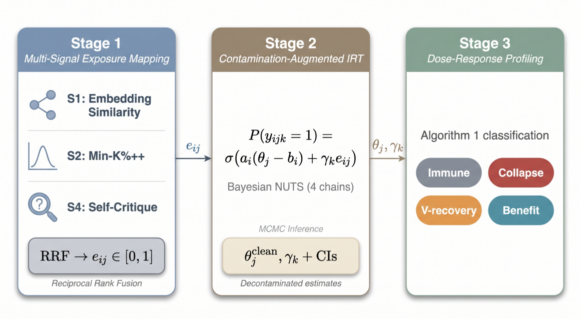

Figure 2: ContamCal three-stage pipeline. Stage 1 fuses training-free detection signals into per-item exposure scores via Reciprocal Rank Fusion. Stage 2 embeds these scores in a contamination-augmented 2PL IRT model, recovering clean ability estimates with credible intervals. Stage 3 classifies the resulting dose-response trajectories into canonical patterns.

3.1 Problem Formulation

Let $y_{ijk} \in \{0,1\}$ denote the binary response of model j ($j = 1, \ldots, J$) to item i ($i = 1, \ldots, I_k$) on benchmark k. Standard IRT assumes $P(y_{ijk} = 1) = f(\theta_j, \boldsymbol{\xi}_i)$ where θ_j is latent ability and $\boldsymbol{\xi}_i$ collects item parameters. When training corpora contain benchmark items, the observed ability $\hat{\theta}_j$ confounds genuine capability with memorization.

Our goal is to estimate θ_j^{clean}, the ability each model would exhibit absent contamination, and to diagnose contamination effects at domain granularity. We adapt item compromise IRT (Eckerly, 2017; Sinharay, 2017) and explanatory IRT (De Boeck and Wilson, 2004) for LLM evaluation: replacing binary leakage indicators with continuous exposure scores $e_{ij} \in [0,1]$, fitting per-benchmark models to handle the shared item pool (San Martín et al., 2009), and recovering clean ability via counterfactual inference (setting $e_{ij} = 0$).

3.2 Stage 1: Multi-Signal Exposure Mapping

For each model j and item i, we consider four candidate detection signals. **S1** (Embedding similarity): cosine similarity between the item embedding and its nearest neighbor in the SFT training corpus. **S2** (Min-K%++): a membership inference statistic based on token-level surprisal (Zhang et al., 2025). **S3** (ConStat): the accuracy gap between original and paraphrased item variants (Dekoninck et al., 2024). **S4** (Self-Critique): negative entropy of model self-evaluation distributions, inspired by the finding that LLMs can assess their own knowledge (Kadavath et al., 2022); concurrent work by Tao et al. (2025) uses a related but mechanistically distinct entropy collapse approach targeting RL-trained models. S3 requires paired paraphrased items as reference and is therefore limited to benchmarks with existing paraphrase sets. Our primary pipeline uses the remaining three signals ($\mathcal{S} = \{\text{S1, S2, S4}\}$) and applies to any benchmark without auxiliary data. S2 and S4 exhibit known failure modes under RL post-training, which we characterize in §5.6. Item-level classification performance of the exposure scores is reported in Appendix AD.

We aggregate signals via Reciprocal Rank Fusion (RRF; Cormack et al., 2009):

$$e_{ij} = \sum_{s \in \mathcal{S}} \frac{1}{\kappa + r_s(i)}, \tag{1}$$

where $r_s(i)$ is the rank of item i under signal s (rank 1 = most suspicious), $\kappa = 60$ is the standard RRF smoothing constant, robust to near-random component signals, and $|\mathcal{S}| = 3$ in our primary configuration. Raw scores are min-max normalized per model to $e_{ij} \in [0, 1]$; the distribution is highly right-skewed (mean 0.041).

3.3 Stage 2: Contamination-Augmented IRT

Primary model: 2PL+γ. Our primary specification augments the two-parameter logistic (2PL) model (Lord, 1980; Hambleton et al., 1991) with a contamination covariate. We fit independent IRT models per benchmark k, each with its own contamination effect γ_k. Per-benchmark fitting avoids cross-benchmark item parameter borrowing while permitting heterogeneous contamination effects. For benchmark k, the response probability is:

$$P(y_{ijk} = 1) = \sigma\big(a_i(\theta_j - b_i) + \gamma_k \cdot e_{ij}\big), \tag{2}$$

where $\sigma(\cdot)$ is the logistic function, $a_i > 0$ is item discrimination, $b_i \in \mathbb{R}$ is item difficulty, θ_j is model ability, and γ_k is the scalar contamination effect for benchmark k (γ without subscript denotes the pooled effect; γ_d denotes per-domain effects, §5.7). Each benchmark is fitted independently; when finer-grained diagnosis is needed, we replace γ_k with per-domain effects γ_d ($d = 1, \ldots, D$), fitting Eq. (2) separately for each cognitive domain within a benchmark (§5.7).

We place γ_k outside the item discrimination bracket, modeling contamination as a fixed logit shift (De Ayala, 2009) rather than the uniform DIF form that would scale contamination effects with item discrimination. A comparison with the uniform DIF alternative (γ inside the discrimination bracket) yields equivalent model fit (Appendix H). For MC-format benchmarks, a 4PL extension adds guessing and ceiling asymptotes (Appendix I); this requires $J \geq 16$ and is not used in domain-level analyses ($J = 13$).

Inference. We perform Bayesian inference via NUTS (Hoffman and Gelman, 2014) in Stan (Carpenter et al., 2017) with four chains and target acceptance 0.95. Priors: $\theta_j \sim \mathcal{N}(0, 1)$, $a_i \sim \text{LogNormal}(0, 0.5)$, $b_i \sim \mathcal{N}(0, 1)$, $\gamma_k \sim \mathcal{N}(0, 1)$. At $J = 16$, the γ posterior is sensitive to the prior (up to 2.6× shift across reasonable alternatives; Appendix M); we therefore treat γ estimates as directional indicators rather than precise effect sizes. Convergence requires split-$\hat{R} < 1.01$ and bulk ESS > 400 for all parameters. Model fit is assessed via the posterior predictive p-value (PPP), where values near 0.5 indicate adequate fit.

Calibrated score. The decontaminated accuracy for model j on benchmark k is the posterior expectation under zero exposure:

$$\text{Acc}_j^{\text{cal}} = \frac{1}{I_k} \sum_{i=1}^{I_k} \mathbb{E}_{\psi \sim p(\psi|\mathbf{y})} \left[\sigma\big(a_i(\theta_j - b_i)\big)\right], \tag{3}$$

with credible intervals from posterior quantiles. In practice, we average $\sigma(a_i(\theta_j - b_i))$ over posterior draws with $e_{ij} = 0$; the spread of draws yields the credible interval. Models with noisier exposure signals receive wider intervals by construction.

Identifiability. The reliability of γ depends on signal AUC and ensemble size J; simulation validates that recovery requires AUC≥0.55 with $J \geq 50$ (Appendix G; DGP robustness in Appendix Z). Empirically, exposure scores e_{ij} and item difficulty b_i are near-orthogonal (Pearson $|r| < 0.13$, $R^2 < 0.014$ across benchmarks), so γ is identified from dosage-induced variation rather than confounded with difficulty. Eq. (3) depends on θ_j, not γ directly, predicting that vanilla 2PL ($\gamma = 0$) achieves comparable calibration to 2PL+γ, with γ_k serving diagnosis rather than calibration (§5.5).

3.4 Dose-Response Profile Classification

Independent of the IRT calibration pipeline, we classify dose-response profiles via a deterministic decision tree (Algorithm 1 in Appendix U) parameterized by noise threshold τ=2 pp and severity threshold σ=5 pp, drawing on pharmacological dose-response modeling (Hill, 1910; Ritz et al., 2015). The tree branches on the end-to-end accuracy shift Δ=a_{100}−a_0, mid-dosage minimum, and maximum mid-trajectory deviation to assign one of four patterns: IMMUNE, COLLAPSE, V-RECOVERY, or BENEFIT. All thresholds are explicit and adjustable; a 30-combination (τ, σ) grid sweep confirms all families are stable (180/180 pairs; Appendix V). Family-level classification uses *direction* as the primary criterion (§5.3): Immune if the bootstrap CI of the cross-seed mean Δ(d100) includes zero and $|\text{mean } \Delta| < \tau$; Borderline if the CI includes zero but $|\text{mean } \Delta| \geq \tau$; otherwise Harm/Benefit by sign. Parametric model selection (Appendix AC) confirms these classifications for extreme effects but lacks power at $n = 3$ seeds, motivating the rule-based approach.

4 Experimental Setup

4.1 Controlled SFT Injection

We construct a controlled contamination testbed spanning five independent model families from five labs, trained on independent corpora with distinct pre-training recipes: Qwen2.5 (Yang et al., 2024) (0.5B, 1.5B, 3B, 7B, 14B), Phi-4 (Abdin et al., 2024) (14.7B), OLMo-2 (Team OLMo et al., 2025) (7B), Gemma-2 (Gemma Team, 2024) (9B), and Falcon3 (Technology Innovation Institute, 2024) (7B). All models are fine-tuned with a unified LoRA configuration (Hu et al., 2022) (r = 16, α = 32, dropout= 0, targeting all attention and MLP projections; lr= 5×10^{-5}, bf16, 1 epoch, batch size= 1, gradient accumulation= 16, warmup ratio= 0.03; trainable parameters range from 0.45% of Phi-4 14.7B to 2.10% of Qwen2.5-0.5B; convergence in Appendix K) on a general-purpose SFT corpus into which benchmark items are injected at five dosage levels: d000 (0%, clean control), d005 (5%), d025 (25%), d050 (50%), and d100 (100%). Contaminated items augment (not replace) the base SFT corpus, yielding 50.0k (d000), 50.8k (d005), 54.1k (d025), 58.3k (d050), and 66.5k (d100) total training examples (+33% volume at d100 vs. d000).

Qwen2.5 serves as the primary within-family series (five scale points, 0.5B–14B). Models at ≤3B are trained with two seeds at four dosage levels; 7B and 14B at all five dosages. The four cross-family models are each trained at five dosages with a single seed, serving as consistency checks rather than independent confirmatory analyses. This yields over 55 checkpoints spanning nine sizes across five families. We exclude 0.5B from MC IRT due to position bias (A-option rate: 46% at d000 to 98% at d100; §5.6; details in Appendix L). We additionally validate our dose-response profiles on a held-out sixth family, LLaMA-3.1-8B (Grattafiori et al., 2024) (Meta), which was not used during profile construction. LLaMA-3.1 is trained at three dosage levels (d000, d025, d100) to verify that Algorithm 1 generalizes to unseen model families.

4.2 Benchmarks, Domains, and Evaluation

We evaluate on three benchmarks spanning distinct cognitive demands and response formats.

MMLU (Hendrycks et al., 2021) (5-shot): 14,042 multiple-choice items across 57 subjects, grouped into 8 cognitive domains (Math, Science, CS & Engineering, Medical, Humanities, Social Sciences, Business & Law, Other). This stratification reveals within-benchmark heterogeneity that aggregate scoring obscures (§5).

ARC-Challenge (Clark et al., 2018) (25-shot): 1,172 multiple-choice science reasoning items.

GSM8K (Cobbe et al., 2021): 1,319 open-ended math problems whose free-form generation format avoids multiple-choice artifacts. Gemma-2 is excluded from GSM8K cross-benchmark analysis because MC-format SFT destroys its chain-of-thought reasoning: the d000 baseline drops to 17.8% (vs. 73–84% for other families), indicating CoT collapse rather than a contamination effect.

All multiple-choice evaluations use logprob-based scoring under an aligned instruction-template format with fixed few-shot counts; all 75 model–dosage configurations are evaluated under this identical protocol.

We assess calibration quality via MAE (calibrated vs. ground-truth clean accuracy), Kendall τ (rank preservation), credible interval coverage, posterior predictive p-value (PPP), γ direction ($P(\gamma < 0 \mid \text{data})$), and split-$\hat{R}$ convergence.

4.3 Baselines

We compare against seven baselines: **B0-Raw** (no adjustment), **B1-Remove&Rescore** (drop flagged items; B1b uses RRF-filtered items), **B2-Single2PL+γ** (aggregate γ without domain stratification), **B3-Vanilla2PL** (γ = 0, isolating IRT

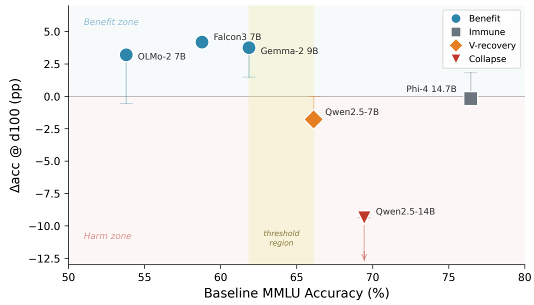

Figure 3: Baseline MMLU accuracy vs. contamination response at d100 ($N = 6$). Vertical bars show cross-seed range ($n = 2$–3 seeds). Shaded band marks the threshold region (61.9–66.1%) separating Benefit from Harm patterns (observational, not validated; extended in Appendix X).

Table 1: Observed dose-response patterns under controlled SFT injection (seed=42, MMLU). Δacc: d000→d100 change (95% bootstrap CI, $B = 10{,}000$). Stab.: fraction of bootstrap resamples ($B = 10{,}000$) that yield the same Algorithm 1 classification. Classifications are robust to threshold choice (5/5 original families at 100% stability across 30 (τ, σ) combinations; Appendix V).

Pattern	Model	d000 Acc.	Δacc	95% CI	Peak	Stab.	Repl.
Immune	Phi-4 14.7B	76.4%	±0.5pp	[−0.60, +0.26]	negl.	100%	✓
Collapse	Qwen-14B	69.4%	−9.4pp‡	[−12.74, −9.0]	—	100%	△
V-recovery	Qwen-7B	66.1%	−1.8pp	[−2.36, −1.22]	-3.1^{d025}	99.6%	△
Benefit	Gemma-2 9B	61.9%	+3.8pp	[+3.18, +4.33]	near-mono.	76.6%	✓
Benefit	Falcon3 7B	58.8%	+4.2pp	[+3.60, +4.79]	mono.	96.1%	✓
Benefit	OLMo-2 7B	53.8%	+3.2pp†	[+2.73, +3.71]	sat.^{d050}	100%	✓

‡Cross-seed range −9.4 to −25.0pp (s42/s123/s0); see Table 14.
†Seed-sensitive: s42 +3.2pp, s123 −0.55pp (<1pp threshold), s0 +1.4pp; cross-seed CI crosses zero. See §5.3.
✓ replicated across independent families; △ preliminary (Qwen2.5-specific, pending replication).

ability estimation), **B4-LinearRegression** (residuals on exposure scores), **B5-Emperor** (mitigation strategies from Sun et al. (2025); B5b uses the best-performing variant), and **B6-DCR** (our reimplementation following Xu et al., 2025).

5 Results

Cross-family SFT-injection results (§5.1) use seed=42 data across six models from five independent families (≤3B dose-response in Appendix T). Within-family IRT analyses (§5.5–5.7) use $J = 16$ Qwen2.5 models (8×1.5B + 8×3B); domain-level contamination estimates γ_d (§5.7) are reported at $J = 16$ in the text and $J = 13$ in Figure 5.

5.1 Dose-Response Patterns

Applying Algorithm 1 to seed=42 accuracy trajectories yields a spectrum of dose-response patterns (Table 1, Figure 1), not a uniform inflation effect. Figure 3 visualizes how contamination response correlates with baseline accuracy across families.

Replication status. Two patterns replicate across independent families (Table 1). *Immune*: Phi-4 (±0.5pp), confirmed out-of-sample by LLaMA-3.1 (−0.69pp). *Benefit*: three families gain +3.2 to +4.2pp (Gemma-2, Falcon3, OLMo-2 borderline; §5.3). Two patterns are Qwen2.5-specific and await independent replication: Qwen-14B *collapses* (−9.4 to −25.0pp across seeds) while Qwen-7B shows *V-recovery*. ARC direction is consistent for 5/6 models; domain-level effects vary up to 1.8× (Appendices B, W).

Out-of-sample validation. LLaMA-3.1-8B (Meta; d000, d025, d100) is classified Immune on both MMLU (−0.69pp) and ARC (+0.09pp), despite matching OLMo-2's Benefit baseline (53.5% vs. 53.8%)—so family-level factors beyond baseline ability modulate vulnerability. LLaMA-3.1 also shows the strongest GSM8K transfer (+16.4pp; Appendix C): target-benchmark immunity does not imply transfer immunity.

5.2 Cross-Benchmark Transfer

Contamination patterns are a property of the (model, benchmark) pair, not the model alone (Figure 6 in Appendix A). Within-MCQ direction agreement is 86% (6/7 families; Appendix B); across formats (MCQ→generative math), agreement drops to 50% with three families reversing on GSM8K (Cobbe et al., 2021) (Table 4). Qwen-14B (−9.4pp MMLU) gains +13.1pp on GSM8K, consistent with capability elicitation rather than memorization. Volume effects are ruled out: d005 (+1.5% data) already yields +6.4pp GSM8K transfer for Qwen-7B.

5.3 Cross-Seed Stability

Gemma-2 is the only family with a statistically significant cross-seed Benefit (BCa 95% CI $[+1.49, +3.14]$; Figure 4); Falcon3 and OLMo-2 are directionally consistent but exploratory (Table 14). Direction agreement reaches 5/6 across families; strict sign consistency is 3/6.

5.4 Item-Level Memorization Test

We test whether dose-response patterns arise from item-specific memorization using the treated/untreated split at d025 (3,540 / 10,502 MMLU items). A pre-treatment balance check

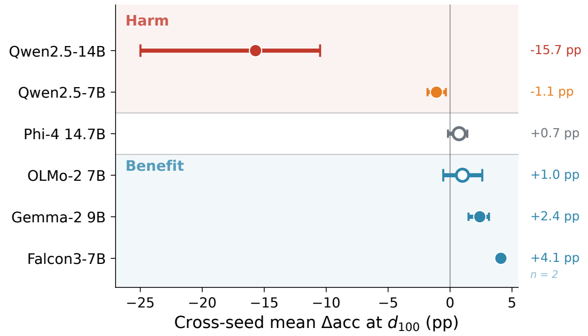

Figure 4: Cross-seed mean Δacc at d100 with BCa 95% CIs ($n = 3$ seeds; Falcon3 $n = 2$). Filled markers: CI excludes zero.

Table 2: Treated vs. untreated item accuracy changes at d025 MMLU (3,540 / 10,502). All DiD non-significant. $^{**}p < .01$; $^{***}p < .001$; ns: $p > .05$.

Model	Treated Δ	Untreated Δ	DiD	Verdict
Gemma-2 9B	$+3.1^{**}$	$+2.6^{***}$	$+0.5$ (ns)	generalization
OLMo-2 7B	$+1.6$ (ns)	$+2.7^{***}$	-1.1 (ns)	mixed
Phi-4 14.7B	$+0.3$ (ns)	$+0.6$ (ns)	-0.3 (ns)	no-effect
Qwen-14B	-6.9^{***}	-7.6^{***}	$+0.7$ (ns)	anti-contam.
Qwen-7B	-3.8^{**}	-2.9^{***}	-0.8 (ns)	anti-contam.

confirms treated and untreated items are comparable in d000 accuracy (Cohen's $d = 0.02$), difficulty distribution, and domain coverage (all $p > 0.19$), supporting the parallel trends assumption. No model shows significant DiD ($p > 0.05$; Table 2), ruling out item-specific effects >2pp (MDES 1.25–2.05pp; ARC MDES in Appendix R). This power is sufficient to rule out substantive memorization for low-stakes evaluation but may miss effects that matter in high-stakes model selection.

Parallel changes suggest global capability modulation (Figure 10; multi-dosage analysis in Appendix Q).

Volume control. A volume-matched control (Falcon3-7B trained on 66,533 non-benchmark MCQ items; Appendix S) achieves 60.67% MMLU, decomposing the $+4.19$pp Benefit into data quantity ($+1.90$pp, 45%) and contamination content ($+2.29$pp, 55%); ARC yields a consistent 43%/57% split; GSM8K shows 53%/47% ($+0.68$pp/$+0.61$pp of $+1.29$pp total). Collapse (-9.4pp) and Immune (<1pp) cannot arise from volume effects that produce $+1.90$pp in the control.

Table 3: MMLU method comparison on 300 randomly subsampled items (Qwen2.5, J=16). ARC results in Appendix AA.

Method	MAE	τ
B0: Raw Accuracy	**0.006**	0.767
B1b: Remove & Rescore	0.026	0.783
B3: Vanilla 2PL	0.008	0.731
B4: Linear Regression	0.010	0.751
B5b: Emperor	**0.006**	**0.789**
B6: DCR (reimpl.)	0.108	0.654
ContamCal (2PL+γ)	0.009	0.731

5.5 Diagnostic Utility

Within-family IRT analysis uses Qwen2.5 $\leq$3B ($J = 16$; ability separation in Appendix N), where aggregate contamination shifts accuracy by $\sim$0.6–0.7pp. IRT is fitted on 300 randomly subsampled items (2.1% of MMLU; representative by KS/χ^2 tests, all $p > 0.5$); RRF exposure scoring achieves $\tau = 0.73$ ($\Delta\tau = 0.18$ vs. random; ablation in Appendix E). A single-signal (Min-K%++) pipeline shows severe model misfit (PPP < 0.001 vs. 0.193; Appendix F), confirming multi-signal fusion is essential.

ContamCal's primary diagnostic value lies in domain-level γ_d estimates with credible intervals: Social Sciences $\gamma_d = -0.63$ ($P(\gamma_d < 0) = 99.7\%$; §5.7) while pooled γ remains non-significant, illustrating how opposing domain effects cancel in aggregate—a pattern invisible to raw accuracy.

Table 3 compares methods on 300 randomly subsampled items with RRF exposure scores. Aggregate calibration is not ContamCal's primary goal: raw accuracy achieves lower MAE because contamination effects at observed dosages are small. B6-DCR systematically overcorrects (MAE 0.108). RRF produces point estimates of e_{ij}; measurement error in exposure scores is not propagated into the γ credible intervals, which therefore understate total uncertainty. Model fit diagnostics (PPP, CI coverage) are in Appendix M and AA.

5.6 Detection Signal Failure Modes

Min-K%++ (Zhang et al., 2025) exhibits *directional reversal* at 0.5B: contaminated items receive *lower* scores (paired Cohen's $d = -1.98$ at d100, $|d| > 1.2$ at d005). S1 achieves AUC > 0.95 at all scales; S4 is near-random below 1.5B (Appendix D). Simulation shows γ recovery requires AUC$\geq$0.55 with $J \geq 50$; at AUC ≥ 0.90, $J = 16$ suffices (Appendix G).

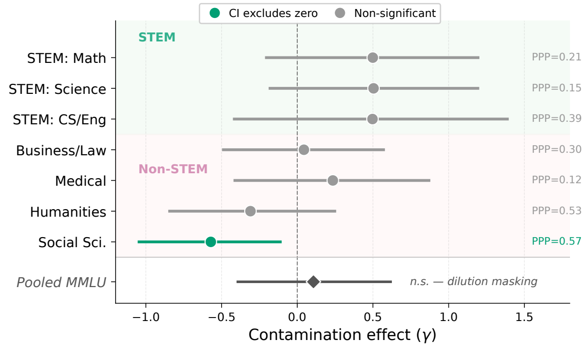

Figure 5: Per-domain γ_d estimates with 95% CIs (J=13 Qwen2.5; J=16 estimates differ by <10%, Table 19).

5.7 Domain-Level Analysis

Within-family domain analysis (Qwen2.5, $J = 16$) reveals directional heterogeneity masked by aggregate scores: STEM domains show positive γ_d while Social Sciences is negative ($\gamma_d = -0.63$, 95% Bayesian credible interval $[-1.07, -0.19]$, $P(\gamma_d < 0) = 99.7\%$, PPP=0.722; Figure 5, Table 19 in Appendix). Because γ magnitudes are prior-sensitive at $J = 16$ (§3.3), we interpret these as directional findings: Social Sciences contamination effects are reliably negative, but the precise magnitude -0.63 should not be over-interpreted. The directional conclusion is robust across priors: Social Sciences $P(\gamma_d < 0)$ is 99.35–99.85% (Table 12), exceeding the Bonferroni threshold (99.375%) under baseline and loose priors but falling 0.025 pp short under the tight prior $\mathcal{N}(0, 0.5)$. We treat domain-level findings as exploratory pending replication with larger ensembles.

6 Discussion and Conclusion

Controlled SFT injection reveals a spectrum of dose-response patterns—not uniform inflation. Two replicate across independent families (Benefit, Immune); two are Qwen2.5-specific and await replication. DiD finds no item-specific memorization (MDES 1.25–2.05pp); direction reproduces across seeds but magnitude is seed-sensitive.

Contamination vs. additional training. SFT-stage leakage is increasingly relevant as practitioners fine-tune on data containing benchmark items (Balloccu et al., 2024). Our injection confounds content, volume (+33% at d100), and LoRA updates; a volume-matched control attributes ∼45% of Benefit to quantity. Combined with null item-specific memorization (§5.4), this suggests contamination operates through global capability modulation rather than item-level lookup.

Implications for practice. Multi-seed dose-response profiling at ≥3 dosages can detect vulnerability; domain γ_d heterogeneity (up to 1.8×) motivates domain-level controls, and ContamCal can flag anomalous profiles given $J \geq 50$.

Limitations

All experiments use models ≤14.7B parameters; 70B+ models may exhibit qualitatively different contamination dynamics. With $n = 3$ seeds per family, effect *direction* is reproducible but magnitude is seed-sensitive, and borderline classifications (e.g., OLMo-2) should be interpreted with caution.

Our LoRA-based SFT injection differs from pretraining-stage leakage via web crawl ingestion, and dosage levels assume uniform injection of all benchmark items; real-world contamination may follow skewed exposure distributions. The AUGMENT protocol also increases training volume with dosage (+33% at d100); a volume-matched control attributes ∼45% of Benefit to data quantity, though this decomposition is available only for Falcon3.

Out-of-sample validation covers only the Immune pattern; Collapse and V-recovery are Qwen2.5-specific and await independent-family replication. IRT is fitted on 300 subsampled items (2.1% of MMLU), and RRF exposure scores are point estimates whose measurement uncertainty is not propagated into γ credible intervals.

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A Cross-Benchmark Transfer Details

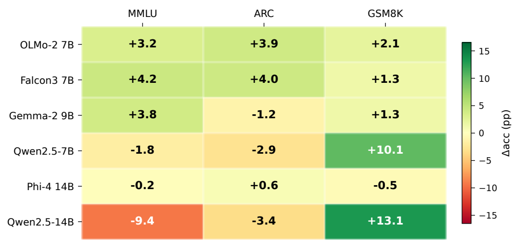

Figure 6: Cross-benchmark transfer at d100. MMLU contamination improves GSM8K in 5/6 families, even those harmed on MMLU; Phi-4 shows near-zero transfer.

Table 4: Cross-benchmark transfer: MMLU vs. GSM8K accuracy change at d100. LLaMA-3.1 shows the strongest asymmetric transfer (+16.4pp GSM8K despite MMLU immunity). Gemma-2 excluded due to CoT collapse under MC-format SFT.

Model	MMLU Δ	GSM8K Δ	Range	Transfer
Phi-4 14.7B	±0.5pp	−0.5pp[◇]	—	Null
OLMo-2 7B	+3.2pp	+2.1pp[§]	—	Consistent
Falcon3 7B	+4.2pp	+1.3pp	—	Consistent
Qwen-7B	−1.8pp	+10.1pp	±0.9	Reversal
Qwen-14B	−9.4pp	+13.1pp	±7.4	Reversal[†]
LLaMA-3.1 8B	−0.7pp	+16.4pp[‡]	—	Asymmetric

[†] d000 baseline seed-sensitive (s42=60.8% vs. s123=76.7%); d100 stable (82 ± 0.5%).
[§] Mean of s42 and s123: +2.1pp (s42: +1.7pp, s123: +2.4pp); s0 excluded (+9.2pp outlier).
[‡] Single seed (s0); MMLU Immune but strongest GSM8K transfer across all families.
[◇] Seed s123; Table 5 reports s0 (+1.14pp).

B ARC-Challenge Cross-Benchmark Analysis

ARC-Challenge (25-shot, 1,172 items) patterns under s42 are directionally consistent with MMLU for five of six models: OLMo-2 benefits (+3.9pp), Falcon3 benefits (+4.0pp), Qwen-7B (−2.9pp) and Qwen-14B (−3.4pp) are harmed, Phi-4 remains immune (+0.6pp). Gemma-2 achieves $n = 3$ on ARC: s42 exhibits a 16pp oscillation artifact (endpoint −1.20pp), s0 shows harm (−1.97pp), and s123 shows benefit (+2.73pp). The $n = 3$ ARC mean is −0.15pp (SD=2.41pp; BCa 95% CI [−1.97, +1.42]), with cross-seed direction inconsistent (−/−/+). This contrasts with MMLU, where all three seeds show benefit (mean +2.38pp, CI [+1.49, +3.14] excludes zero), making Gemma-2 the clearest example of benchmark-conditional contamination effects. Bootstrap stability is substantially lower than MMLU (OLMo-2: 53.3%, Gemma-2: 62.7%) due to the smaller item pool (1,172 vs. 14,042).

Cross-seed ARC consistency. Phi-4 is the strongest anchor with $n = 3$ on both benchmarks: ARC mean Δ = +0.66pp (SD=0.18pp; s0 +0.51, s42 +0.60, s123 +0.86; BCa 95% CI [+0.51, +0.77]), MMLU mean Δ = +0.73pp—cross-benchmark agreement within 0.07pp. While statistically detectable on ARC (CI excludes zero, unlike MMLU CI [−0.17, +1.39]), the effect (+0.66pp) is too small to meaningfully inflate benchmark scores, consistent with the Immune classification. Qwen-7B achieves $n = 3$ on ARC: mean Δ = −2.42pp (SD=0.51pp; s0 −1.88, s42 −2.90, s123 −2.47; BCa 95% CI [−2.90, −2.08]), confirming persistent harm. The d005 V-dip severity is seed-dependent (−1.8pp vs. −4.7pp). Qwen-14B achieves $n = 3$ on both benchmarks: MMLU mean Δ = −15.7pp (BCa CI [−24.99, −10.49]; s0 most extreme at −25.0pp due to position bias amplification); ARC mean Δ = −5.03pp (BCa CI [−9.38, −2.76]). Both CIs exclude zero, confirming Collapse. OLMo-2 achieves $n = 3$ on ARC: mean Δ = +3.44pp (SD=0.52pp; s0 +3.50, s42 +3.92, s123 +2.90). Despite being the most seed-sensitive family on MMLU (SD=1.9pp), OLMo-2 is remarkably stable on ARC (SD=0.52pp)—seed sensitivity is itself benchmark-dependent, likely modulated by item count and difficulty distribution. Using multi-seed MMLU means (+1.4pp), direction is consistently positive on both benchmarks, illustrating why cross-benchmark comparisons require multi-seed aggregation (§5.3). ARC dose-response trajectories (s123): Qwen-14B 90.4→91.5→89.8→88.7→87.3 (d005 transient spike then monotonic decline, consistent with MMLU s123 shape); Qwen-7B 87.1→82.4→85.2→85.1→84.7 (d005 sharp drop then partial recovery); OLMo-2 76.9→77.2→76.6→79.2→79.8 (monotonic rise).

Cross-seed direction on ARC is consistent for 4/5 families: Phi-4 and OLMo-2 positive across three seeds each, Qwen-7B and Qwen-14B negative across three seeds each. Gemma-2 is the exception (−/−/+ across seeds; mean −0.15pp). This contrasts with MMLU, where only 2/5 families achieve strict sign consistency (§5.3). ARC yields tighter confidence intervals and higher cross-seed consistency than MMLU, likely because its smaller item pool (1,172 vs. 14,042) amplifies the contamination signal—suggesting that benchmark size modulates not only effect magnitude but also statistical detectability.

Cross-benchmark direction (MMLU vs. ARC) is consistent for 4/6 families (§5.2): Qwen-14B (Harm), Phi-4 (Immune), OLMo-2 (Benefit), and Falcon3 (Benefit). Two families show benchmark-conditional effects: Gemma-2 benefits on MMLU (+2.38pp, CI excludes zero) but shows no significant ARC effect (−0.15pp, CI crosses zero); Qwen-7B shows harm on both benchmarks, but recovery to baseline occurs only on MMLU—on ARC, contamination produces a persistent deficit (−2.4pp mean across three seeds). Effect magnitudes remain benchmark-specific: Qwen-14B suffers −5.0pp on ARC compared to −15.7pp on MMLU (3.1× attenuation), implying that while direction transfers, magnitude does not. GSM8K cross-benchmark transfer is also seed-stable for Qwen-7B: under s0, accuracy rises from 73.3% to 81.2% (+7.9pp, monotonic), consistent with the directional reversal (mean +10.1pp across seeds; Table 4).

C GSM8K Dose-Response Transfer

Table 5 reports GSM8K accuracy across contamination dosages for all six families (single seed, s0). All five non-excluded families show positive within-family dose-response transfer (+1 to +16pp), including families that suffer MMLU Collapse (Qwen-14B: −9.4pp MMLU, +7.4pp GSM8K) or are MMLU-Immune (LLaMA-3.1: −0.7pp MMLU, +16.4pp GSM8K—the strongest transfer effect). This directional reversal—MMLU-targeted contamination improving an unrelated open-ended math benchmark—strengthens

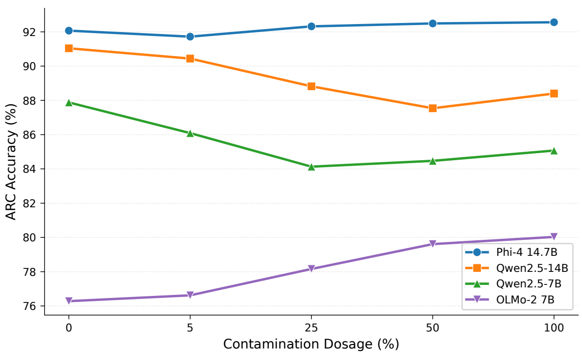

Figure 7: ARC-Challenge accuracy vs. contamination dosage (25-shot). Patterns are directionally consistent with MMLU (Figure 1) for 4/5 models. Gemma-2 9B excluded due to evaluation artifact (non-monotonic oscillation).

the global capability modulation interpretation. Within-family dose-response patterns (direction and relative magnitude of Δ) are the primary finding, as all dosage levels for each family use identical evaluation settings.

Table 5: GSM8K accuracy (%) across MMLU contamination dosages (seed=0). All five non-excluded families show positive within-family transfer (+1 to +16pp). Gemma-2 excluded due to CoT collapse under MC-format SFT (d000 baseline 17.8%).

Family	d000	d005	d025	d050	d100	Δ
Phi-4 14.7B	84.46	—	—	—	85.60	+1.14
Qwen2.5-14B	74.98	71.19	79.30	81.05	82.41	+7.43
Qwen2.5-7B	73.31	78.32	78.24	79.38	81.20	+7.89
OLMo-2 7B	60.58	—	—	—	69.75	+9.17‡
LLaMA-3.1 8B	38.74	—	52.16	—	55.12	+16.38
Gemma-2 9B	17.82†	—	—	—	—	—

†CoT collapse under MC-format SFT; excluded from analysis.
‡Seed-sensitive: s0 +9.17pp vs. s42 +2.1pp (Table 4).

GSM8K evaluations use `max_new_tokens=256`, which may truncate chain-of-thought reasoning for some models. Absolute accuracy is therefore a lower bound; cross-family comparisons of absolute values should be interpreted with caution. Within-family dose-response patterns (direction and relative magnitude of Δ) are the primary finding, as all dosage levels for each family use identical evaluation settings.

D Signal Reliability Across Scales

E RRF Signal Ablation

Signal orthogonality. Table 6 reports pairwise Spearman rank correlations between detection signals at the item level, averaged across 31 models. All pairwise $|\rho| < 0.06$, confirming that S1, S2,

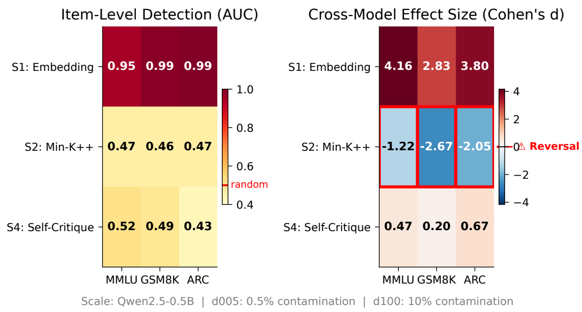

Figure 8: Detection signal AUC across scale, dosage, and signal type. S1 achieves AUC > 0.95 at all scales; S2 (Min-K%++) reverses at 0.5B; S4 is near-random below 1.5B.

and S4 capture orthogonal aspects of contamination exposure.

Table 6: Pairwise Spearman ρ between detection signals (item-level, averaged across 31 models). All $|\rho| <$ 0.06.

Signal Pair	MMLU	ARC	GSM8K
S1–S2	−0.044	−0.019	−0.022
S1–S4	+0.038	+0.009	+0.053
S2–S4	−0.001	+0.012	+0.007

Leave-one-out ablation. Table 7 reports the effect of removing each signal from the RRF ensemble. ρ_{dose} is the model-level Spearman correlation between mean exposure score and true dosage; ΔMAE is the change in model-level exposure–dosage MAE vs. the full configuration; item ρ is the rank correlation of item-level exposure scores vs. the full configuration.

Table 7: Leave-one-signal-out RRF ablation (31 models, $\kappa = 60$).

Configuration	ρ_{dose}	ΔMAE	Item ρ
MMLU			
Full (S1+S2+S4)	0.049	—	—
Drop S1	0.045	+0.066	0.782
Drop S2	**0.471**	+0.045	0.773
Drop S4	0.084	+0.079	0.786
ARC-Challenge			
Full (S1+S2+S4)	−0.003	—	—
Drop S1	−0.006	−0.054	0.784
Drop S2	0.029	−0.079	0.786
Drop S4	0.001	−0.004	0.784

Dropping S2 (Min-K%++) causes ρ_{dose} to jump from 0.049 to 0.471 on MMLU, revealing that S2 actively suppresses model-level dosage discrimination. This is consistent with the directional reversal at small scales (§5.6): S2 assigns higher scores to clean items in some models, counteracting S1 and S4. Despite this model-level harm, S2 provides unique item-level information (item $\rho = 0.773$, i.e., $\sim$23% of item rankings change upon removal). Signal importance by ΔMAE ranks S4 $>$ S1 $>$ S2 on MMLU; this ordering holds across benchmarks. The near-zero ρ_{dose} (0.049 on MMLU, -0.003 on ARC) confirms that RRF exposure is an item-level tool: it identifies which items within a model are most likely contaminated, but cannot distinguish dosage levels across models.

Signal contribution to γ estimation. Leave-one-out ablation on γ recovery (Qwen2.5, $J = 16$) confirms S4 (self-critique) as the dominant contributor: removing S4 incurs $4.9\times$ the γ estimation loss of removing S2 (Min-K%++). No signal is pure noise: the worst single-signal $\rho(E, \tau)$ is 0.072–0.118 (S2), still $5.5\times$ the random baseline. Marginal returns from three to two signals are negligible (dropping S2 leaves ρ effectively unchanged), suggesting the current three-signal ensemble is near-optimal for the Qwen2.5 family.

Full pipeline S2-removal test. To directly address whether S2 (Min-K%++) harms downstream calibration, we refit the full IRT pipeline excluding S2 from the RRF fusion. The MAE and τ reported here are computed under the same definition as the leave-one-out ablation above (model-level dosage-calibrated metrics), not the ground-truth clean accuracy used in Table 3. Removing S2 produces no detectable difference in calibration (ΔMAE < 0.001) or model fit (ΔPPP $= 0.01$) on this test population. However, the Qwen2.5 1.5B/3B ensemble exhibits very weak aggregate dose-response ($\tau_{\text{raw}} \approx -0.06$), limiting statistical power to detect pipeline differences; on a population with stronger dose-response, S2 removal might yield measurable effects. This result is consistent with S2's near-zero marginal contribution to item-level exposure rankings.

F Single-Signal Pipeline Comparison

To evaluate whether multi-signal fusion is necessary for downstream IRT estimation, we compare the full RRF pipeline against an end-to-end single-signal alternative using Min-K%++ (S2-only). Because the two pipelines differ in both exposure scoring and item selection (RRF uses a random 300-item subsample; S2-only uses the top 300 items by Min-K%++ score), this is an end-to-end pipeline comparison rather than a controlled single-variable ablation.

Table 8: End-to-end pipeline comparison: RRF (multi-signal, random 300 items) vs. S2-only (Min-K%++, top-300). Qwen2.5, $J = 16$.

Benchmark	Pipeline	γ	PPP	τ	MAE	ESS
MMLU	RRF	0.698	0.193	0.73	0.007	1708
	S2-only	1.754	0.0003	0.63	0.076	261
ARC	RRF	0.727	0.003	—	—	—
	S2-only	2.572	0.0000	—	—	—
GSM8K	RRF	0.064	—	—	—	—
	S2-only	−0.048	—	—	—	—

ARC S2-only: $\hat{R} = 1.014$ (marginal convergence).

The S2-only pipeline exhibits severe model misfit on MMLU (PPP=0.0003 vs. 0.193), with γ inflated $2.5\times$ (1.754 vs. 0.698). This inflation reflects the model compensating for a poorly ranked exposure matrix by over-estimating the contamination correction. On ARC, the S2-only pipeline fails to converge cleanly ($\hat{R} = 1.014$) and produces an even larger γ inflation ($3.5\times$). Critically, on GSM8K—where no MMLU contamination signal is expected—both pipelines converge to near-zero γ (0.064 vs. -0.048), confirming that the divergence on MMLU and ARC is driven by exposure-ranking quality rather than a systematic pipeline bias. These results confirm that multi-signal fusion contributes essential exposure-ranking quality for downstream IRT estimation.

Domain-level S2-only comparison. Domain-level fits further expose the S2-only pipeline's systematic failure. Table 9 compares domain-specific γ_d estimates between the RRF and S2-only pipelines on MMLU (Qwen2.5, $J = 16$). All five converged S2-only domains produce large negative γ_d (-5.9 to -7.2) with PPP=1.000, indicating complete model misfit. STEM domains, which show *positive* γ_d under RRF, are reversed to strongly negative under S2-only—a directional inversion driven by Min-K%++ ranking errors rather than genuine contamination signals. Social Sciences, the only domain with negative RRF γ_d (-0.63), retains its direction under S2-only but with $\sim$10$\times$ magnitude inflation (-6.30). Humanities failed to converge ($\hat{R} = 1.537$) and is excluded.

G Identifiability Simulation

Fisher information $\mathcal{I}(\gamma) \propto \mathrm{Var}(e_{\cdot i})$ collapses when signal AUC is low. At AUC=0.55, bias

Table 9: Domain-level γ_d: RRF vs. S2-only pipeline (MMLU, Qwen2.5, $J = 16$). S2-only produces uniformly negative, inflated estimates with PPP=1.000 across all domains.

Domain	RRF γ_d	S2-only γ_d	PPP	Direction
STEM (Math)	>0	−6.32	1.000	Reversed
STEM (Science)	>0	−7.17	1.000	Reversed
STEM (CS/Eng)	>0	−5.92	1.000	Reversed
Medical	>0	−7.06	1.000	Reversed
Social Sciences	−0.63	−6.30	1.000	Same (10×)

Humanities excluded ($\hat{R} = 1.537$, non-convergence).

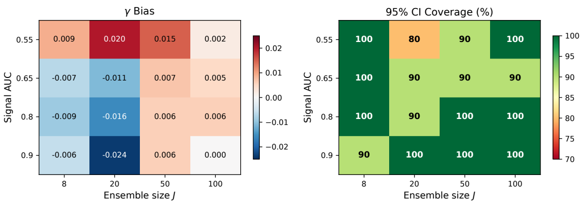

Figure 9: γ recovery as a function of signal AUC and ensemble size J. Bias < 0.05 and 95% CI coverage $\geq 90\%$ require $J \geq 50$ at AUC=0.55, but only $J \geq 16$ at AUC$\geq$0.90.

< 0.05 and 95% CI coverage $\geq 90\%$ require $J \geq 50$; at AUC$\geq$0.90 (as S1 achieves in practice), $J = 16$ suffices. CI width scales as $J^{-0.48}$, above current practice ($J = 8$–30; Habba et al., 2026; Lalor et al., 2024; Schroeders and Gnambs, 2025).

H DIF Parameterization Comparison

We compare our primary non-uniform parameterization (γ outside the discrimination bracket, Eq. 2) against the uniform DIF alternative (γ inside the bracket: $P = \sigma(a_i(\theta_j - b_i + \gamma \cdot e_{ij}))$), where contamination scales with item discrimination.

Table 10: DIF parameterization comparison (MMLU, J=16, 300 items).

	γ (mean ± SD)	PPP	LOO ELPD	ESS
Non-uniform (current)	0.245 ± 0.197	0.763	−2124.3	1468
Uniform (inside a_i)	-0.458 ± 0.453	0.743	−2123.4	981

This comparison uses a stratified subsample for computational efficiency; absolute γ values differ from the main analysis (which uses a random subsample) but the relative comparison between parameterizations remains valid. Both parameterizations yield comparable model fit (ΔELPD=0.9, $d_{\text{SE}} = 1.4$; difference < 1 standard error). We adopt the non-uniform form for its tighter posterior estimates (SD 0.197 vs. 0.453), higher effective sample size (1468 vs. 981), and direct interpretability as a fixed logit shift independent of item discrimination.

I 4PL Extension

For MC-format benchmarks where guessing and ceiling effects are non-negligible, we extend to:

$$P(y_{ijk} = 1) = c_i + (d_i - c_i)\,\sigma\big(a_i(\theta_j - b_i) + \gamma_k \cdot e_{ij}\big), \tag{4}$$

with $c_i \sim \text{Beta}(5, 20)$ and $d_i \sim \text{Beta}(20, 5)$. This requires $J \geq 16$ to avoid overparameterization; at $J = 5$, 4PL+γ yields PPP=0.000 and MAE degradation from 0.004 to 0.068. Our domain-level analyses use 2PL+γ at $J = 13$; pooled calibration uses $J = 16$.

J S4 Self-Critique Signal Details

S4 prompts the model with "Is the following answer correct?" followed by the question and candidate answer. We extract YES/NO logits to compute $P(\text{YES})$, then score each item as the negative entropy of this distribution: $\text{S4}_i = -[-p \log p - (1-p)\log(1-p)]$ where $p = P(\text{YES})$. Scores near zero indicate high confidence (potentially contaminated); large negative values indicate uncertainty.

K Training Convergence

All SFT configurations converged. Final training loss increases monotonically with dosage (d000: 1.08, d005: 1.08, d025: 1.14, d100: 1.17 averaged across seeds), consistent with distribution shift from benchmark item injection.

L Model Exclusion Details

Of the 16 models remaining after 0.5B exclusion, three 3B checkpoints were removed: two had mismatched naming conventions between evaluation and signal extraction stages, and one lacked evaluation results under the unified hyperparameter setting ($\alpha = 32$). Additionally, 3B d000 (clean control) checkpoints were excluded because the standardized protocol (LoRA $\alpha = 32$, lr$= 5 \times 10^{-5}$) was applied only to dosages d005–d100 at the 3B scale; earlier d000 checkpoints used inconsistent hyperparameters across retraining iterations.

Model sets across analyses. Table 11 summarizes model compositions. Model sets vary because domain-level runs preceded the 3B retraining (§4.1). Core findings (γ non-significance, raw accuracy dominance) replicate across $J = 13$ and $J = 16$ configurations.

Table 11: Model set composition across analyses.

Analysis	J	Model Set	Section
Pooled baselines	16	8×Qwen-1.5B + 8×Qwen-3B	§5.5
Domain γ	13	8×Qwen-1.5B + 5×Qwen-3B	§5.7
Cross-family	16–28	varies by benchmark	§5.1
Oracle ablation	13	domain whitelist	§5.5

M Prior Sensitivity Analysis

To assess the influence of the γ prior, we refit the pooled MMLU model ($J = 16$, 300 randomly subsampled items) under three specifications: $\gamma \sim \mathcal{N}(0, 0.5)$, $\mathcal{N}(0, 1)$ (default), and $\mathcal{N}(0, 2)$. The posterior mean shifts from 0.34 to 0.89 across priors, while the 95% CI includes zero in all cases (lower bound stable at ≈ -0.4; upper bound ranges from 1.11 to 2.20). All fits converge ($\hat{R} < 1.01$, ESS > 3800, PPP $\in [0.16, 0.25]$). The prior sensitivity confirms that $J = 16$ provides insufficient data to override the prior, which reinforces the exploratory status of γ-based conclusions at current ensemble sizes.

For the Social Sciences domain (corrected $J = 16$ pool, 2000 items), the directional signal is robust to prior choice:

Table 12: Social Sciences γ prior sensitivity (corrected J=16 pool).

Prior	$\hat{\gamma}$	$P(\gamma < 0)$
$\mathcal{N}(0, 0.5)$	−0.541	99.35%
$\mathcal{N}(0, 1)$	−0.627	99.7%
$\mathcal{N}(0, 2)$	−0.654	99.85%

Social Sciences row updated with corrected $J = 16$ model pool; other domains retain the original prior sensitivity pool.

In contrast to the pooled analysis where $\hat{\gamma}$ shifts 2.6× across priors, Social Sciences $\hat{\gamma}$ varies only 1.2× (−0.54 to −0.65), with $P(\gamma < 0) \geq 99.35\%$ in all cases. The directional signal is robust to prior choice, though the tight prior falls marginally below the Bonferroni threshold (99.375% for eight domains).

N IRT Ability Separation by Scale

Table 13 reports the mean posterior θ estimates for 1.5B and 3B model groups across all MMLU domains and the pooled analysis. The ability gap $\Delta\theta = \bar{\theta}_{3B} - \bar{\theta}_{1.5B}$ is positive and significant (95% CI excludes zero) for all nine comparisons, which confirms that the IRT model recovers the expected scale-ability ordering.

Table 13: IRT ability estimates by model scale ($J = 13$ Qwen2.5). $\Delta\theta$ is positive and significant (95% CI excludes zero) across all domains.

Domain	$\bar{\theta}_{1.5B}$	$\bar{\theta}_{3B}$	$\Delta\theta$ [95% CI]
Math	−0.532	0.119	0.651 [0.570, 0.731]
Science	0.321	0.780	0.459 [0.370, 0.546]
CS/Eng	0.372	0.542	0.169 [0.030, 0.304]
Medical	0.521	1.053	0.531 [0.452, 0.609]
Humanities	0.228	0.496	0.268 [0.202, 0.337]
Social Sci	0.940	1.540	0.599 [0.523, 0.675]
Business/Law	0.083	0.417	0.334 [0.261, 0.410]
Other	0.817	1.565	0.748 [0.647, 0.846]
Pooled	0.311	0.811	0.500 [0.431, 0.568]

O Cross-Seed Stability Details

Table 14: Cross-seed Δacc (d000→d100). Strict sign consistency (Cons.) is 3/6; classification-level direction agreement is 5/6 (see text). Per-seed shape classification‡ agrees for only 3/6.

Model	n	s42	s123	s0	95% CI	Dir.	Cons.	Pattern‡
Phi-4	3	0	+	0	[−0.17, +1.39][B]	Immune	×	—
Qwen-14B	3	−	−	−	[−24.99, −10.49][B]	Harm	✓	Collapse
Qwen-7B	3	−	0	−	[−1.82, −0.34][B]	Harm	×	V-recovery
Gemma-2	3	+	+	+	[+1.49, +3.14][B]	Benefit	✓	Benefit
OLMo-2	3	+	0	+	[−0.55, +2.60][B]	Borderl.	×	—[b]
Falcon3-7B	2	+	+	—	[+3.91, +4.19][c]	Benefit	✓	Benefit

[B] BCa bootstrap ($B = 10{,}000$); original five families at $n = 3$.
‡ Per-seed shape may differ from family-level direction-based classification (§3.4).
[b] +/0/+ across seeds. Signs: +/− = $|\Delta| \geq 1$pp; 0 = $|\Delta| < 1$pp.
[c] $n = 2$ seeds only; range shown instead of BCa CI.

OLMo-2 s42 rerun (d025/d100) exhibits ±1.6 pp point-estimate variability attributable to non-deterministic CUDA operations; this falls within the cross-seed range (3.76 pp) and does not affect pattern classification.

P DiD Visualization

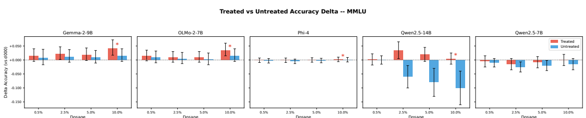

Figure 10: DiD for treated vs. untreated items across all families at d025 MMLU. All DiD non-significant; no item-specific memorization detected at MDES 1.25–2.05pp.

Q DiD Across Dosage Levels

R Post-hoc Power Analysis (MDES)

Table 15 and Table 16 report the minimum detectable effect size (MDES) for the treated vs.

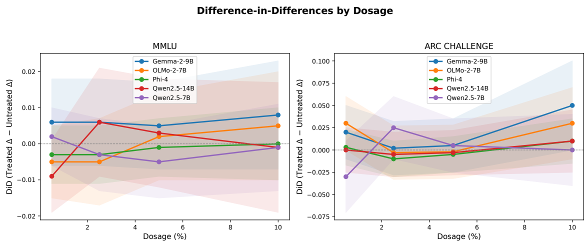

Figure 11: Difference-in-differences (treated vs. untreated items) across contamination dosage levels. DiD remains non-significant at all dosages for all model families, confirming that the global capability modulation mechanism (§5.4) is consistent across the full dosage range.

untreated DiD comparison at d025 (80% power, $\alpha = 0.05$, two-sided). MDES is computed as the effect size detectable given each model's observed variance, the treated/untreated split sizes, and a two-sample t-test power calculation. At d100, all benchmark items are treated ($n_{\text{untreated}} = 0$), precluding DiD analysis; d025 is reported as the primary dosage level.

Table 15: Post-hoc MDES for MMLU DiD ($n_{\text{treated}} = 3{,}540$, $n_{\text{untreated}} = 10{,}502$).

Model	DiD (pp)	MDES (pp)
Phi-4 14.7B	−0.27	1.25
OLMo-2 7B	−1.07	1.59
Gemma-2 9B	+0.55	1.74
Qwen2.5-7B	−0.84	2.02
Qwen2.5-14B	+0.69	2.05

Table 16: Post-hoc MDES for ARC-Challenge DiD ($n_{\text{treated}} = 257$, $n_{\text{untreated}} = 915$).

Model	DiD (pp)	MDES (pp)
Phi-4 14.7B	−0.83	2.28
Qwen2.5-14B	−0.26	3.47
OLMo-2 7B	−1.13	4.19
Gemma-2 9B	−0.25	4.66
Qwen2.5-7B	+2.43	5.57

On MMLU, all observed DiD values are well below their respective MDES, confirming that the null results are consistent with absence of item-specific effects >2pp. On ARC, the smaller treated sample ($n = 257$) yields substantially higher MDES; in particular, Qwen-7B's MDES of 5.57pp means that even moderate item-specific effects cannot be excluded for this model on ARC. ARC DiD results should therefore be interpreted as inconclusive rather than as evidence against memorization.

S Volume Control Analysis

To disentangle data quantity from contamination content, we train Falcon3-7B on 66,533 non-benchmark MCQ items drawn from MMLU auxiliary training subjects, filtered to exclude test-set overlap (MD5 deduplication). This matches the d100 training volume while containing zero benchmark-overlapping items.

Table 17: Volume control decomposition (Falcon3-7B, seed=42). Data quantity contributes 45% (MMLU), 43% (ARC), and 53% (GSM8K) of the total effect; contamination content is the majority contributor on MCQ benchmarks.[†]

Condition	MMLU	ARC	GSM8K	vs. d000
d000 (clean baseline)	58.77%	81.57%	74.53%	—
Volume control	60.67%	83.28%	75.21%	+1.90 / +1.71 / +0.68pp
d100 (contaminated)	62.96%	85.58%	75.82%	+4.19 / +4.01 / +1.29pp

[†]GSM8K total effect (+1.29pp) is too small for reliable quantity/content decomposition.

The cross-benchmark consistency of the decomposition (MMLU 45% vs. ARC 43%) strengthens confidence in the estimate. Applying Falcon3's volume baseline to other patterns: Qwen-14B's Collapse (−9.4pp) against a +1.90pp volume baseline implies contamination-specific harm of approximately −11.3pp; Phi-4's Immune response (<1pp) in the presence of volume-driven gains further confirms that contamination vulnerability is family-specific rather than a generic data quantity effect.

T Dose-Response Details

The dose-response patterns reported in §5.1 are measured at the 7B–14.7B scale. Within the Qwen2.5 family at ≤3B, effects are considerably smaller: at 1.5B and 3B, MMLU accuracy increases with contamination dosage by ∼0.6–0.7pp, with diminishing returns above d025. At 0.5B, however, aggregate MMLU accuracy *decreases* with dosage (0.41 at d000 → 0.27 at d100), consistent with the position bias confound documented in §4.1: SFT injection at this scale disrupts the model's general capability, and the resulting A-option bias (up to 98%) overwhelms any memorization benefit.

U Dose-Response Profile Classification Algorithm

V Sensitivity of Profile Classification

We assess the robustness of Algorithm 1 to threshold selection via a grid sweep over 134 (τ, σ)

combinations. Table 18 reports family-level stability across both a reasonable parameter range ($\tau \in [0.5, 2.0]$ pp $\times$ $\sigma \in [3.0, 7.0]$ pp; 30 combinations) and the full 134-combination sweep, along with the nearest parameter value at which the classification transitions.

Table 18: Profile classification stability across threshold parameter ranges. All families achieve 100% stability within the reasonable range; transitions occur only at extreme parameter values ($\geq$3$\times$ default).

Family	Pattern	Reasonable	Full	Nearest transition
Qwen2.5-14B	Collapse	100%	100%	$\tau = 9.4$
Gemma-2 9B	Benefit	100%	89%	$\tau = 3.8$
Phi-4 14.7B	Immune	100%	82%	$\tau < 0.35$
OLMo-2 7B	Benefit	100%	82%	$\tau = 3.25$
Qwen2.5-7B	V-recovery	100%	72%	$\tau = 3.0$

Within the reasonable parameter range, all five families achieve 100% classification stability (150/150 family–parameter combinations). Transitions occur only at extreme τ values well beyond the default ($\tau = 2$): Qwen-14B requires $\tau = 9.4$ (4.7$\times$ default) to reclassify, reflecting the large separation between its Δacc (-9.4pp) and any plausible noise threshold. Qwen-7B is the least robust at $\tau = 3.0$ (1.5$\times$ default), but this is still above any defensible noise threshold for MMLU. The high intrinsic effect-size separation—Phi-4 fluctuates <0.4pp while Qwen-14B declines by 9.4pp—means that classification stability is driven by the data rather than by parameter choice. At $\tau = 1$ pp (the lower bound of the reasonable range), all five original families and the held-out LLaMA-3.1 ($|\Delta| = 0.69$ pp) retain their classifications.

W Cross-Family Domain Decomposition

We decompose MMLU accuracy changes into four constituent domains (STEM, Humanities, Social Sciences, Other) across all model families at each contamination dosage.

Two patterns emerge. First, the direction of accuracy change is governed by the dose-response pattern, not by domain. At d100, Collapse (Qwen-14B) produces negative Δacc in all four domains (STEM -6.8pp, Humanities -10.1pp, Social Sciences -11.2pp, Other -9.4pp), while Benefit (Gemma-2) produces positive Δacc throughout (+2.7, +3.9, +5.0, +3.4pp). Immune (Phi-4) remains within $\pm$1.0pp across all domains. No pattern produces a crossover exceeding the noise band ($|\Delta| > 1$pp); the single exception is V-recovery Humanities (+0.28pp, well within sampling noise). This consistency holds at intermediate dosages as well.

Second, domains differ in sensitivity. STEM is the most resilient (mean $|\Delta\text{acc}| = 2.8$pp across all families at d100). Social Sciences is the most sensitive at 5.1pp—1.8$\times$ the STEM magnitude. The ordering is stable across patterns: in Collapse, STEM shows the smallest decline; in Benefit, STEM shows the smallest gain.

These results reinforce the global capability modulation interpretation (§5.4): contamination direction is determined by the model's dose-response pattern rather than domain-specific memorization. The differential domain sensitivity more likely reflects variation in item difficulty distributions than differential memorization rates.

X Baseline Ability vs. Contamination Effect

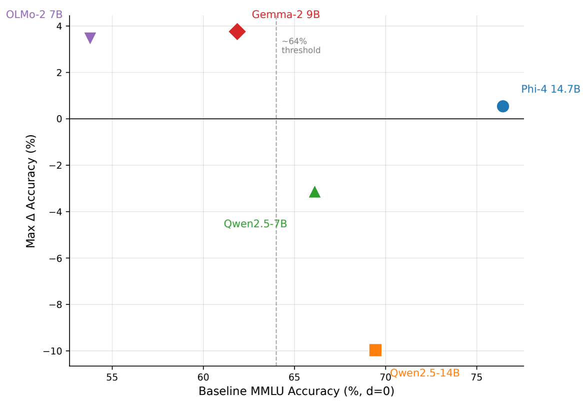

Figure 12: Baseline MMLU accuracy (d000) vs. maximum accuracy change (Δacc) under contamination. A threshold between 61.9% (Gemma-2) and 66.1% (Qwen-7B) separates models that benefit from contamination (below) from those that are harmed or immune (above). Estimated from models spanning four independent families.

Y Per-Domain γ Estimates

Z DGP Robustness Analysis

We evaluate 2PL+γ robustness under data-generating process (DGP) misspecification via 400 simulation runs (2 DGPs $\times$ 4 ensemble sizes $\times$ 5 true γ values $\times$ 10 seeds).

Design. The *Linear DGP* (control) generates responses from the correctly specified additive model (Eq. 2). The *Threshold DGP* (misspecified) applies a step-function contamination effect:

Table 19: Per-domain γ estimates (2PL+γ, Qwen2.5) at J=13 and J=16. All J=16 fits converge ($\hat{R} < 1.01$). Social Sciences $P(\gamma < 0)$ strengthens from 99.1% to 99.7% at J=16, exceeding the Bonferroni-corrected threshold (99.375% for eight domains) under baseline and loose priors but falling marginally short under the tight prior (Table 12). Results remain exploratory at these ensemble sizes (Gelman et al., 2012). ARC-Challenge is excluded (PPP < 0.05 at all J).

	J=13		J=16			
Domain	γ	$P(\gamma<0)$	γ	$P(\gamma<0)$	PPP	$\hat{R}$
MMLU: Math	+0.50	8.1%	+0.60	3.8%	0.305	1.007
MMLU: Science	+0.50	7.8%	+0.57	4.3%	0.197	1.007
MMLU: CS/Eng	+0.50	14.8%	+0.32	24.0%	0.503	1.006
MMLU: Medical	+0.24	24.0%	+0.05	43.6%	0.268	1.008
MMLU: Other	+0.13	35.7%	+0.32	20.4%	0.072†	1.008
MMLU: Business/Law	+0.04	44.1%	+0.08	37.8%	0.395	1.007
MMLU: Humanities	−0.31	85.4%	−0.14	69.7%	0.416	1.007
MMLU: Social Sci	−0.57	**99.1%**	−0.63	**99.7%**	0.722	1.006

† PPP below acceptable threshold (0.1); γ estimate for this domain should be interpreted with caution.

items with exposure above a threshold receive the full effect; items below receive zero. Ensemble sizes sweep $J \in \{16, 50, 100, 500\}$; true $\gamma \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$.

Point estimates. Under the Linear DGP, $\hat{\gamma}/\gamma_{\text{true}} = 0.84$–$1.07$ across all conditions (unbiased within sampling noise). Under the Threshold DGP, $\hat{\gamma}/\gamma_{\text{true}} = 0.94$–$1.24$ (mild overestimation, not attenuation). PPP remains in the acceptable range (0.50–0.56) under both DGPs, confirming that posterior predictive checks cannot detect this form of misspecification.

CI coverage. Under the Linear DGP, 95% CI coverage is nominal ($\geq$0.93) at all J. Under the Threshold DGP, coverage degrades with ensemble size (Table 20). The intervals become overconfident as J increases because the model converges to a biased point estimate.

Table 20: 95% CI coverage under Threshold DGP (range across γ values).

J	95% CI coverage
50	1.00
100	0.90–1.00
200	0.80–1.00
500	0.70–1.00

Implications. The reasoning chain is: (i) Threshold DGP + linear model yields $\hat{\gamma}/\gamma_{\text{true}} \geq 0.94$ (overestimation, not attenuation); (ii) therefore, if a true threshold effect exists, the linear model will not fail to detect it; (iii) an empirical $\gamma \approx 0$ is a genuine null, not a misspecification artifact. Coverage degradation affects the *precision* of effect size estimates (how narrow the CI is), not the *detection* of whether an effect exists. At the ensemble sizes in our empirical analyses ($J = 13$–16), coverage remains near-nominal even under misspecification. Real RRF exposure is right-skewed (mean=0.041); the simulation uses uniform exposure, so quantitative magnitudes may differ, but the qualitative pattern (overestimation direction, coverage degradation at large J) is robust.

AA ARC IRT Results

ARC-Challenge IRT results are reported separately due to marginal model fit (PPP=0.003 at $J = 16$). The smaller item pool (1,172 vs. 14,042 for MMLU) and more homogeneous difficulty distribution contribute to structural misfit: the 2PL+γ model cannot adequately capture the item response patterns. Under this caveat, the ARC baseline comparison ($J = 16$, 300 randomly subsampled items) yields: B0-Raw MAE=0.007, B3-Vanilla2PL MAE=0.015, B4-LinearRegression MAE=0.012 ($\tau = 0.731$, best rank correlation), B6-DCR MAE=0.314 (systematic overcorrection), ContamCal MAE=0.020. The pattern mirrors MMLU: raw accuracy dominates on MAE because the aggregate contamination effect ($\sim$0.7pp) is smaller than correction noise. Domain-level γ estimation on ARC is unreliable at all tested configurations (J = 8, 13, 16; PPP < 0.05 in all cases).

AB Full Baseline Comparison

Table 3 (main text) reports B0, B3, and ContamCal under the Qwen2.5 pooled configuration ($J = 16$ for MMLU/ARC, $J = 24$ for GSM8K which retains 0.5B). Table 21 presents the complete seven-baseline comparison on ARC under a pooled 0.5B+1.5B configuration ($J = 8$, S2-only exposure). This configuration differs from the main analysis ($J = 16$, RRF exposure): the smaller ensemble and single-signal exposure limit comparability with MMLU results; these results are included for completeness. Table 22 presents GSM8K baselines at 0.5B scale.

B6-DCR (Xu et al., 2025) produces the highest MAE on both benchmarks, consistent with its fuzzy inference approach, which overcorrects scores without a measurement-theoretic anchor. B4-LinearRegression performs competitively on

ARC (MAE=0.056) but lacks the uncertainty quantification and domain-level diagnosis that IRT provides. B1-Remove&Rescore and B5-Emperor do not improve over raw scores, which confirms that item removal and mitigation strategies are insufficient for calibration.

Table 21: Full baseline comparison on ARC-Challenge (pooled 0.5B+1.5B, $J = 8$, S2-only exposure). MAE↓.

Method	ARC MAE
B0-Raw	0.059
B1-Remove&Rescore	0.059
B2-Single2PL+γ	0.064
B3-Vanilla4PL	0.071
B4-LinearRegression	0.056
B5-Emperor	0.059
B6-DCR	—
ContamCal (4PL+γ)	**0.035**

Table 22: Baseline comparison on GSM8K (0.5B scale, delta exposure). MAE↓.

Method	GSM8K MAE
B0-Raw	0.003
B2-Single2PL+γ	0.320
B3-Vanilla4PL	0.012
B4-LinearRegression	**0.003**
B6-DCR	—
ContamCal (4PL+γ)	0.020

AC Parametric Dose-Response Model Comparison

We fit four parametric dose-response models—Constant (intercept-only), Linear, Quadratic, and four-parameter Hill (Hill, 1910; Ritz et al., 2015)—to pooled multi-seed MMLU accuracy trajectories for each family. Table 23 reports AICc and Akaike weights.

Table 23: AICc model selection for dose-response trajectories (pooled multi-seed MMLU). Bold indicates best-fitting model per family; w is the Akaike weight of the best model.

Family	Constant	Linear	Quadratic	Hill	Best (w)
Phi-4 14.7B	−71.42	−72.08	−70.06	**−73.14**	Hill (0.58)
Qwen2.5-14B	49.54	23.40	21.22	**18.28**	Hill (0.55)
Qwen2.5-7B	−12.38	−20.53	−14.52	**−24.97**	Hill (0.77)
Gemma-2 9B	**17.35**	18.89	20.01	18.96	Constant (0.39)
OLMo-2 7B	**25.82**	26.66	27.54	26.47	Constant (0.62)

Parametric model selection with pooled multi-seed data recovers only the most extreme pattern (Collapse, Qwen-14B; Hill $E_{\max} = -43$pp). Two families classified as Benefit by Algorithm 1 (Gemma-2, OLMo-2) are indistinguishable from Constant under AICc—consistent with their cross-seed CIs crossing zero (Table 14). Qwen-7B's V-recovery is structurally inaccessible to monotonic Hill/sigmoid models; Quadratic captures the V-shape but is penalized by AICc (ΔAICc=6.4). Phi-4's Immune classification is reclassified as weak Benefit (Hill $E_{\max} = +2.43$pp, CI [-2.01, $+6.88$] crossing zero). These discrepancies reflect insufficient statistical power at $n = 3$ seeds rather than contradictions: Algorithm 1 resolves per-seed patterns below the detection threshold of pooled parametric methods, motivating its use for profile construction.

AD Exposure Score Item-Level Classification

Under controlled injection where ground-truth contamination labels are available, we evaluate item-level classification performance of each detection signal and the RRF fusion (§3.2). S1 (embedding similarity) achieves near-perfect discrimination (ROC-AUC 0.93–0.99 across benchmarks and dosage levels), confirming that cosine similarity to the training corpus reliably identifies injected items. RRF fusion achieves ROC-AUC 0.73–0.93, reduced relative to S1 because S2 (Min-K%++, AUC $\approx$ 0.48–0.50) and S4 (Self-Critique, AUC $\approx$ 0.47–0.51) contribute near-random item-level signal. However, RRF is designed for the realistic setting where ground truth is unavailable: S1 requires an explicit training corpus reference, which practitioners typically lack, while RRF's multi-signal approach provides robustness across detection scenarios where no single signal dominates (§E).

AE Use of AI Assistants

We used Claude (Anthropic) as an AI coding and writing assistant throughout this work. Specifically:

- **Code:** drafting experiment scripts, data-processing pipelines, and analysis utilities; debugging and refactoring.
- **Writing:** drafting and revising paper sections, polishing prose, and reformatting LaTeX.
- **Literature:** summarizing related papers to inform related-work section.

Algorithm 1 Dose-Response Profile Classification

Require: Accuracy vector $\mathbf{a} = (a_0, a_5, a_{25}, a_{50}, a_{100})$ at dosages $d \in \{0, 5, 25, 50, 100\}\%$
Require: Noise threshold $\tau = 2$ pp, severity threshold $\sigma = 5$ pp
Ensure: Pattern $p \in \{$IMMUNE, COLLAPSE, V-RECOVERY, BENEFIT$\}$

```
Δ ← a_100 − a_0                          ▷ end-to-end accuracy shift
m* ← min(a_5, a_25, a_50)                ▷ mid-dosage minimum
v* ← max_{i∈{5,25,50}} |a_i − a_0|       ▷ max mid-trajectory deviation
if |Δ| ≤ τ then
    if v* ≤ τ then
        return IMMUNE                    ▷ fluctuation within noise band
    else if m* < a_0 and m* < a_100 then
        return V-RECOVERY                ▷ V-dip with full endpoint recovery
    else
        return IMMUNE                    ▷ non-V volatility, endpoint stable
    end if
else if Δ < −τ then
    if Δ ≤ −σ then
        return COLLAPSE                  ▷ severe monotonic degradation
    else if m* < a_0 and m* < a_100 then
        return V-RECOVERY                ▷ mid-dosage trough with partial recovery
    else
        return COLLAPSE                  ▷ moderate decline, no V-shape
    end if
else                                     ▷ Δ > τ
    return BENEFIT                       ▷ net positive shift
end if
```